

Munesu Manyangarirwa

Data Scientist / Technical Mentor / Program Operations Analyst

Data scientist and program-operations analyst based in Johannesburg, South Africa, working across workforce analytics, applied statistics, and data automation. Background in Industrial & Organisational Psychology, currently completing Honours at UNISA. Four years of continuous, hands-on data work spanning statistical modelling, SQL and BI dashboarding, and Python/Google Cloud automation pipelines.

About & Skills

My day-to-day work sits between program operations and applied data science: building automated analytics pipelines and BI dashboards that program teams rely on daily, while keeping statistical and machine-learning skills active through dedicated projects. This portfolio deliberately shows both sides rather than only the more presentable half.

PROGRAMMING & DATA

Python

SQL

Apps Script

pandas

NumPy

scikit-learn

Git

BI & VISUALISATION

Power BI

DAX

Looker Studio

Matplotlib

Seaborn

CLOUD & AUTOMATION

Google Cloud

Sheets API

Service Accounts

Pipeline Design

ANALYSIS & DOMAIN

Statistical Modelling

Classification

Workforce Analytics

A note on scope. A meaningful share of my recent work, ALX program operations data and confidential financial-sector analysis under NDA, cannot be shown in detail here. Where that applies, the case study describes the method and architecture rather than the underlying data, and no real learner or client data appears anywhere in this document or the accompanying GitHub portfolio.

Four Years, Continuously

JUL 2024 →
PRESENT

Program Operations Associate

ALX Training SA · Remote

- Managed end-to-end operations for two programs across four concurrent cohorts of 1,000+ learners
- Designed automated Apps Script dashboards, cutting manual processing by ~30 min per program per day
- Built graduation trackers that lifted graduation rates by 3-4 percentage points per program

JAN 2024 →
JUN 2024

Data Science & Data Analytics Technical Mentor

ALX Training SA · Remote

- Supported 1,000+ learners across Data Analytics and Data Science programs
- Built a Python-powered Looker Studio pipeline, cutting daily processing from 4 hours to 1.5
- Delivered 10 live webinars; initiated a BigQuery migration proposal for the team's data infrastructure

SEP 2023 →
DEC 2023

Junior Technical Mentor

ALX Training SA · Remote

- Provided technical support to learners and coordinated webinar delivery with Course Leads

JUL 2022 →
OCT 2022

Data Science Intern

ExploreAI · South Africa

- HR statistical modelling, extreme-value analysis, and weather-forecasting models
- Built BI dashboards and visualisations for executive audiences using Python

MAR 2021 →
JUN 2021

Project Lead

North-West University

- Led a cross-disciplinary semester project, improving the project score by 30%

Selected Projects

Learner Engagement Risk Model

Python · scikit-learn · 2026

A predictive-modelling companion to the descriptive engagement segmentation I run operationally: logistic regression and random forest classifiers trained on a synthetic dataset built to mirror real engagement signals (dormancy, quiz completion, activation speed, program stage). Built specifically to keep statistical/ML skills active alongside day-to-day automation work.

- Full evaluation: accuracy, precision, recall, F1, ROC-AUC, and ROC curve comparison
- Class-weight balancing for the minority (churn) class; feature-importance analysis
- No real learner or client data used anywhere in this project

Program Analytics Automation Suite

Apps Script · Python · GCP

An automation architecture replacing manual, spreadsheet-by-spreadsheet tracking across concurrent training programs: a shared Apps Script entry point driving pacing, payment, and archive updates, paired with a Python/Colab pipeline that aggregates larger datasets and pushes results into Sheets and Looker Studio via a service account.

- Cut manual reporting time by ~30 minutes per program per day; adopted beyond its original team
- Shown here as a generalised architecture description; production code is under NDA

Program Pacing & Engagement Analytics

Python · pandas · Seaborn

A recurring analytics pipeline answering a structured set of research questions about learner pacing and engagement: drop-off funnels, activation timing, dormancy risk, and cohort equity checks across concurrently-run programs, refreshed on a recurring cadence for stakeholder reporting.

SQL & BI Work

Maji Ndogo Water Access Analysis

SQL · Power BI

Built and delivered as the technical mentor facilitating ExploreAI/ALX's relational-analysis curriculum: multi-table joins across four raw tables, a reusable analysis view, CTEs calculating province- and town-level access percentages, conditional business logic classifying infrastructure improvements, and a window-function ranking query for engineering prioritisation, then visualised in Power BI with a geographic shape map and KPIs.

HR People Analytics Dashboard

Power BI · DAX

A three-page Power BI report analysing employee absenteeism and productivity: demographic and lifestyle segmentation, absence patterns by reason and weekday, and a workload-vs-absenteeism scatter feeding a KPI gauge and rolled-up productivity score, built with a direct line to my Industrial & Organisational Psychology background.

Power BI Teaching Demo: Company Financials

Power BI

A self-authored teaching dataset and starter dashboard (revenue vs. allocated funds), designed from scratch to teach learner cohorts Power BI fundamentals through a guided, scenario-based exercise. All data is invented for instructional purposes.

2022 Foundations

Python · scikit-learn

Regression, classification, and unsupervised-learning sprint projects from my ExploreAI Data Science internship and certification, including extreme-value analysis applied to weather-event forecasting. The foundation the rest of this portfolio builds on.